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Original article

A novel wireless sensor network deployment for monitoring and predicting abnormal actions in medical environment and patient health state

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ABSTRACT

Wireless Sensor Networks (WSNs) play a crucial role in modern healthcare applications by sensing and communicating environmental and patient health data through wireless mediums. These networks enable real-time monitoring of patients, including the elderly and individuals with chronic conditions, offering significant benefits in surveillance and healthcare management. However, several challenges persist in implementing WSNs for healthcare. Real-time administration faces obstacles such as managing patients, medical infrastructure, and staff while ensuring timely treatment delivery. Moreover, accurately predicting diseases and providing precise treatment remains complex due to the diverse nature of medical data and the limitations of traditional systems. This study proposes a deep learning-based approach integrated with WSNs to address these challenges. The system employs deep convolutional neural networks (DCNNs) and long short-term memory (LSTM) models to automate healthcare functions such as patient monitoring, infrastructure management, and disease prediction. The proposed model (DCNN-LSTM) analyzes diverse data types, including numeric, alphanumeric, images, signals, and video, to accurately identify abnormal conditions and predict diseases. The system also generates real-time alerts for medical personnel, ensuring timely intervention. The proposed framework was validated using NS2 simulation software to model the WSN architecture and sensor node communication. Experimental results demonstrate that the DCNN-LSTM models achieved a classification accuracy of 96 % with a loss rate of 0.08, showcasing significant improvements over traditional approaches. Integrating WSNs and deep learning enhances real-time healthcare monitoring, improving efficiency and patient outcomes.

1. Introduction

Sensors with low power but better technology are used in new fields like medicine. WSNs possess the potential to enhance quality outcomes in numerous applications and enhance excellence across diverse configurations and segments of the population [1–4]. The way WSNs have been utilized in numerous applications is fascinating, such as real-time patient surveillance in hospitals, emergency care in massive catastrophes, enhancing the quality of life for the elderly and research into human and persistent illnesses. Some early prototypes of WSNs have shown that they can help detect when someone's health is worsening [5,6] and provide emergency responders at the earliest [7]. Remote monitoring and controlling such events are done with WSNs, which comprise routers, nodes, and gateways [8]. The sensor nodes monitor, gather, and exchange information between various locations and are classified into actual and sensor nodes. The routers enhance the communication range based on demand, and the gateway is used to control and send data from the nodes to the end user of the real-time applications [9]. WSN integrated systems are extensively used in numerous real-time applications and industries, including but not limited to military, transportation, healthcare, environment, industries, agriculture, and area monitoring [10]. The healthcare industry has been using WSN-based applications to keep track of patients and manage hospitals. Wireless sensor devices have been extensively developed and utilized in numerous healthcare sectors to provide prompt treatment to prevent patients from dying and other physical ailments [11,12]. The body sensors, which can be worn or implanted, are among the most popular wireless healthcare devices. This sensor continuously monitors the daily routines of patients. This WSN-based continuous monitoring

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Nomenclature

N	Number
S	Section
MI	Medical industry
S_i	Different sections in the section
Pa	Patients
K	Total number of patients
P	Patients with sensors
H	Sensor hubs
b^k	Bias value
K^{th}	Kernel's map
(I, j)	Kernel's map position
X	Input image
P_{ij}^k	Pooling layer output generated from the kernel's map
$\max\left(A_{p,q}^k\right)$	Maximum activation function value obtained from the previous layer
s	Size of the pooling
F_i,	The final output of the model on the ith class
A_j	The weight value between the jth to ith class in the model
W	The weight matrix
$\widetilde{c}_t$	Candidate memory
B	Bias
i_t	Input gate
σ	Sigmoid function
f_t	Forget gate
c_t	Current moment memory cell
c_{t-1}	The value for the last LSTM unit
*	Dot product
o_t	Output gate
h_t	Output for the LSTM cell

system helps to detect diseases at an early stage and severity level. An intelligent healthcare monitoring and control system has been created by the convergence of IT, medicine, and engineering [13]. The general structure of a wireless sensor network system is given in Fig. 1.

The WSN system in healthcare has many sensor nodes. Those sensor nodes work together and organize the input process independently. During this phase, the WSN-based system will incur higher energy consumption. Data transmission from the sensor to the user requires security. It is essential to protect and transmit the data without losing any data. Recently, WSN systems based on Artificial Intelligence (AI) have been extensively utilized in numerous real-time applications. AI is a technology that works like the human brain to process. Machine and deep learning models are used to learn and train the data using the AI-based approach. Deep learning is an important technique in WSN-based systems to improve model performance. It's a subset of the machine learning algorithm employed to acquire substantial data from the source. It includes various algorithms such as RNN, CNN, LSTM, DNN, etc. This paper uses a deep convolutional neural network (DCNN) and Long-Short-Term-Memory (LSTM) model to control and monitor patients' health records and give doctors continuous alerts. The subsequent section delves into various previous research studies, the workflow of the proposed model, and its outcomes, and concludes with some highlighted points about the proposed DL-based model.

2. Advantages of WSN in Medical

The wireless sensor network (WSN) contains geographical sensors that can easily monitor environmental status like heat, temperature, pressure, humidity, sound, vibration, pressure, light, object motion, etc. So, it can be used in every industrial area, such as precision agriculture, the automotive industry, and medical monitoring [15]. This WSN is more beneficial in the healthcare industry for monitoring health conditions, in-home assistance, telemedicine, smart nursing homes and wireless body area networks. WSN can provide non-stop monitoring of physical parameters like blood pressure, temperature, heart rate, and non-invasive monitoring of oxygen levels. The WSN monitors the patient at the right time, improves disease management and cost-effectiveness, and helps with diagnosis accuracy. It will make human life more comfortable and expand their life [16].

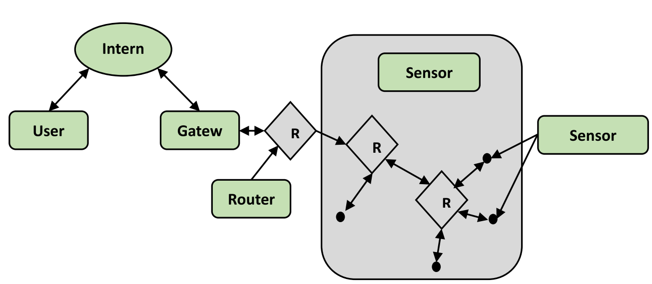


Fig. 1. General structure of WSN [14].

The WSN system in healthcare is made up of multiple sensor nodes. Those sensor nodes work together and self-organize the input process. During this stage, the WSN-based system consumes more energy. In addition, security is vital during data transmission from the sensor to the user. It is essential to protect and transmit the data without any loss. So, in recent years, Artificial intelligence (AI) based WSN systems have been widely used in many real-time applications. AI is a technology that mimics the functions of the human brain to process the data analysis and classification process. The AI-based technique comprises various learning methods, such as machine and deep learning models, to learn and train the data [17].

2.1. Advantages of deep learning in medical

The deep learning (DL) method has become a vital source and is applied and used in various fields. In medical fields, deep learning-based models are used in various tasks, such as classification, image segmentation, disease prediction, and feature extraction. Due to the success of the deep learning model, it is also used to predict overall survival rates. However, deep learning is one of the most popular techniques WSN-based systems use to improve model performance. It is the subset of the machine learning model used to learn a large set of input data. It includes various algorithms such as RNN, CNN, LSTM, DNN, etc. This paper implements deep convolutional neural network DCNN and Long-Short-Term-Memory (LSTM) models to control and monitor patients' health records and provide continuous alerts to doctors.

2.2. Advantage of convolutional neural network (CNN) in medical

DL-based convolutional neural networks (CNN) are a very useful technique in medical image analysis. CNN is commonly captured because of the ability to preserve spatial relationships when filtering the input or given image. To identify the features of the image, CNN mainly uses a feedforward neural network. Only a few units are connected to the fully connected layer; other units in each layer are fully connected to the adjacent layer. CNN has many advantages in this medical field. They are that it speeds up the convergence speed by reducing the parameters effectively, and it can also reduce the parameters by removing the trivial function in the parameters.

2.3. Advantage of long short-term memory (LSTM) in medical

LSTM is also a deep learning model. It also provides several advantages to this medical field using the LSTM model. This model can predict the next stage of the disease by keeping the futuristic and historical information because the LSTM model has forgotten the gates for more data. This model can overcome the shortcomings of other LSTM models by using the backward processing layer in the LSTM model. When it comes to providing accuracy, it is more efficient and powerful than the other normal LSTM model. For predicting more accuracy, it gets the information from the historical data to use it for future purposes.

The main objective of this paper is:

- To present a detailed study of the deep learning model and its uses in medical industries for disease diagnosis.
- To create an efficient deep learning model for predicting different types of diseases from various medical data of the patients.
- To evaluate the performance of the model in terms of performance metrics. Then, the obtained result is compared with other models to prove the model's efficiency.

The following section discusses various earlier research works, the workflow of the proposed model with results, and concludes with some highlighted points regarding the proposed DL-based model.

3. Literature survey

Various research works are traditionally performed to establish a more efficient WSN system. Especially in medical science, multiple authors have suggested various techniques and methods for deploying WSN-based healthcare applications. This section talks about some recent research on WSN-based applications.

3.1. WSN based diseases detection

The authors [16,17] discussed WSN applications used in IoT-based healthcare applications. This IoT system keeps track of patient-related information for the healthcare industry. The WSN-based systems perform better than traditional data-sharing techniques. The dynamic behaviour of WSNs and their sensors can be understood by discussing the performance of the WSNs in healthcare applications [18]. A neural network static random access (NN-SRAM) approach has been proposed to optimize the energy constraint of a WSN. The model saved 76.99 % of energy. A study was done [19] to make the WSN system authentication better. For this reason, a new classification system based on WSNs is proposed, and a comparison is conducted between the existing and current methods. The output shows that the current models provide better output than existing ones. Shankeri et al. [20] proposed a Geographical Information System based optimization algorithm for the WSN system to provide an effective health and environmental monitoring system. Various WSN-based applications will be reviewed and discussed for this first. Second, the lifetime and coverage rate of the WSN are optimized using the MST method. Thirdly, the performance of the proposed mode is assessed by comparing it with the PSO and BEE's algorithm in terms of complexity, convergence rate, and constancy repeatability. On IoT-based WSN systems, the proposed DIS model with the PSO algorithm outperformed the overall analysis result with a high SD of 0.0150. The recent advances in WSN-based systems in healthcare are investigated and presented [21]. The analysis indicates that the WSN-based system has successfully addressed various healthcare limitations. Further, more research has been done to create a WSN-based healthcare system with fewer expenses. An IoT-smart healthcare system based on RFID and WSN was developed [22] for medical data analysis. This model also includes various tools, like Spark, Kafka, NodeJs, and MongoDB. The model's performance is assessed by diagnosing Wolff-Parkinson-White syndrome through logistic regression. The experiment results show that the proposed model can analyze the data with 95 % accuracy.

An IoT-based smart wireless healthcare system has been deployed [23] for continuous patient and healthcare monitoring. This system uses RFID, smartphones, brain sensors, and a headband that senses the human brain. These technologies are incorporated through the IPV6 protocol into a wireless medium. A wearable smart sensor and RFID-based wristwatch monitor the patient's health condition. Patients without a doctor will benefit from this technique.

3.2. WSN based IoT devices for disease detection

An IoT-based smart wireless healthcare system has been deployed [23] for continuous patient and healthcare monitoring. This system uses RFID, smartphones, brain sensors, and a headband that senses the human brain. These technologies are incorporated through the IPV6 protocol into a wireless medium. A wearable smart sensor and RFID-based wristwatch monitor the patient's health condition. A patient without a doctor will benefit from this technique. Kaidi et al. [24] presented a detailed study that defined various IoT-based sensors and techniques used in healthcare units. The IoT features are connected with architecture and explored with the components of Healthcare Monitoring Systems for monitoring the patient's health condition. It uses different kinds of IoT sensors to perform various tasks, and aggregating and recovering all the data is crucial. Thus, they aimed to use a separate LPWAN with WiFi for data aggregation and transmission speedily and effectively. After data transmission, it is essential to analyze and predict health conditions. E. Aarthi et al. [25] have proposed an ML-based model to identify early cardiac symptoms with the help of Wireless Sensors Networks (WSN) and IoT. It was also used to identify and classify patients' health conditions immediately. The main aim of this paper is to connect the IoT device with WSN to improve the monitoring system in the healthcare industry, and then it will improve the detection and help people reduce the risk for heart disease. The NBC method was used to analyze the data, and it outperformed other methods like BS-DNN and CNN. The comparison shows that the proposed Naïve Bayes to achieve a % classification accuracy of 96 % is better than the different methods. Recently, Wireless Sensor Networks based on the Internet of Things have been highly used in healthcare applications because IoT networks can effectively generate and transmit data and provide more benefits like energy, security, and QoS in data management. Murthy, M. Y. B., & Koteswararao, A. [26] have used IoT in WSN applications to improve the overall QoS in the network. The author, Krishnamoorthy et al. [27], defined an intelligent healthcare monitoring system based on new technologies like the IoT and machine learning. The proposed IoT-ONN model has obtained an accuracy advantage of 4–15 % over conventional methods. They are considering the IoT-ONN model for ANNs trained with the BPN method, saving the training time from 15 % to 52 %, and through the output, it was described that the proposed model provided more accuracy in the training phase.

Ardila and Vivares-Builes [28] have proposed a PRISMA-based methodology to manage the database in WSN applications. It was focused on finding and detecting cavities, face mask applications, toothbrush forces, and physiological parameters from dental implants using wireless sensors. Using sensors can reduce the workload of the dental practitioner, whereas the sensor data needs to be analyzed immediately, but this has not yet been done. Saini et al. [29] used BAN logic, ProVerif, and other WSN-based tools for mutual authenticating and security verification used in the proposed protocol in WSN. All the sensors used in the WSN are IoT sensors, and they perform well when combined with security tools.

3.3. Machine learning model-based diseases detection

The author H. Sharma et al. [30] presented how ML algorithms can be used in WSN-based smart cities. The machine learning model is an optimized tool for WSN-IoT applications in smart cities. The supervised learning algorithm performs better and is widely used compared to unsupervised and reinforcement learning models. The computational time and cost are reduced by an ML-based model for HIoT [31–36]. A congestion-rerouting strategy boosts the model's efficiency. The proposed model is made up of two parts. In the first step, the ML algorithm is used to predict the size of the packet forwarding window. In the second step, data packets are prioritized and sent to the predicted window size. The comprehensive analysis indicates that the proposed model is more efficacious for the Healthcare Internet of Things. Awotunde et al. proposed an ML-based Io-WBN framework for smart healthcare sectors in 2021. The wearable sensors gather input data, including heartbeat rate, glucose level, and body temperature. An AI algorithm is applied to extract the essential features from the input data. This model is widely used to monitor patient health conditions remotely. This research primarily focuses on providing regular monitoring, reducing costs, minimizing medical faults, and enhancing patient satisfaction. M. Pundir and J.K. Sandhu [35] and G. Gardasevic et al. [36] reviewed the quality-of-service parameters of the ML algorithm applied to the WSN system. It examines the data from 2011 to 2021 to determine the model's efficiency—a few suggestions for future researchers to boost its performance. Aldabbas et al. [37] propose the implementation of a machine learning and clustering-based algorithm for diagnosing appropriate medication for patients. In this case, the spark technique manages the input data. The k-means clustering algorithm is used to extract features and classify diseases. Finally, NB and RF models are used to classify drugs for patients. The overall analysis shows that the NB and RF models accurately classified drugs 95.52 % and 99.23 %.

3.4. Deep-learning based diseases detection

Elbagoury et al. [38] have introduced the hybrid GMDH and LSTM deep learning models for mobile health edge computing using IoMT. It is mainly designed for detecting the heart–stroke disease. This paper uses AI algorithms for stroke prediction and diagnosis. The CNN and GMDH are used for prediction, and LSTM deep learning enhances biomedical signal prediction. Electromyography data signals identify normal and abnormal actions in stroke cases. CNN has proven 98 % accuracy for stroke diagnosis. GMDH monitors the EMG signal for patient health prediction with a normal accuracy of 98.60 % to a minimum accuracy of 96.68 %. Finally, the GMDH and LSTM models reach the normal prediction level of up to 99 %. Lavanya et al. [39] have proposed preprocessing and estimation of deep learning methods that could be done in the edge-computing layer. Using this deep learning (DL) technique, acute stroke prediction can be obtained by analysing physiological data and heart rate variability features. The DL-based wearable sensors of edge computing methods can perform better than the existing work. Finally, the performance of the DL approach is combined with DWT for better performance of acute stroke prediction and estimate the cardiac arrhythmia, including their sensitivity up to 99.4 %, accuracy up to 99.3 and specificity up to 99.1 %.

Elbagoury et al. [40] have proposed a deep learning model for Stroke prediction EMG signal using the Inception-v3 architecture and LSTM model. This study compares the Inception-v3 of Mobile AI with the existing Mobile AI stroke engine, which is a combination for predicting EMG signals with the help of a hybrid LSTM model. Both models performed well, but inception-v3 has 98 % of average accuracy. Jin et al. [41] have implemented the data fusion process that can be used to develop the fusion tensor from multiple data sensors after BiLSTM may perform it to improve the data fusion mechanism. It is mainly used for stroke prediction and accurately identifying the patient's activities. This process is used to observe the long-term dependencies and temporal patterns and enhance wearable sensor-based activity through classification. Compared to the existing system, the proposed models performed well. Lai et al. [42] have introduced an explainable DL model for addressing microwave imaging in multiple object localisation. DL has implemented the gradient-weighted class activation map (Grad-CAM) and Delay-Multiply-And-Sum (DMAS) algorithm to predict the spatial localisation of microwave head strokes. This Grad-CAM technique identifies and shows the important features in the input signal that could be used for decision-making and shown in the real image domain for a better understanding of the output model. Rajkumar, G. et al. [43] has proposed that heart disease is the leading cause of death worldwide. Predicting it is complex and requires expert doctors. Recently, IoT technology has emerged to predict diseases using sensor data accurately. This system proposes a deep learning framework to predict heart disease using a public Hungarian dataset. The data is cleaned and processed using Median Studentized Residual and Harris Hawk Optimization. Then, Modified Deep Long Short-Term Memory (MDLSTM) classifies the data into normal and abnormal. The results are implemented in Python and evaluated using metrics like accuracy, sensitivity, and execution time. The results show that this method is superior in predicting heart disease with 98.01 % accuracy and a reduced error rate of 91.11 %, outperforming existing techniques.

Ge, B. et al. [44] This paper propose that heart sounds can reveal the health of your heart. A recording of these sounds is called a phonocardiogram (PCG). Pulmonary hypertension with congenital heart disease (CHD-PAH) is a severe heart condition that can lead to disability and death. This study aimed to develop a computer-aided diagnosis to detect CHD-PAH early on. We used a non-invasive and simple method that analyses a single heart cycle with multiple features. Here's how it works as we pre-processed the heart sounds to segment each cardiac cycle, we extracted features from the time-frequency domain, wavelet packet energy, and cardiac cycle, we used a convolutional neural network (CNN) to extract deeper features, we combined these features into a single vector, we used XG Boost to classify the heart sounds into normal, CHD, or CHD-PAH and, we used a majority voting algorithm to get the best result from multiple cardiac cycles. Our method achieved an accuracy of 88.61 %. This could be a hopeful tool for early diagnosis of CHD-PAH. Deepa Priya, B. S et al. [45] The rise in air pollution has led to an alarming increase in lung disease deaths. Early detection is key to saving lives, and deep learning can help. By analyzing medical images like X-rays and CT scans, deep learning algorithms can assist doctors in diagnosing lung cancer sooner. This research explores the most effective deep-learning techniques for lung disease prediction. We evaluated various models and measured their performance using standard metrics. Our goal is to harness the power of automation to enhance medical diagnosis and treatment.

Luo, L et al. [46] Breast cancer has reached alarming rates worldwide, making early detection and treatment crucial. Breast imaging is a vital tool in this fight. Deep learning technology has made remarkable strides in analysing breast cancer images, offering new hope for improved diagnosis and treatment. This paper reviews the past decade's progress in deep learning-based breast cancer imaging research, covering various imaging modalities. We explore the main methods and applications, highlighting challenges and opportunities for future research. By addressing these challenges, we can harness the full potential of deep learning to improve breast cancer outcomes. Pedada, K. R. et al. [47] proposed a model for measuring brain tumour size accurately, which is vital for effective treatment. MRI scans help, but manual analysis is slow and requires expertise. we developed a computer-aided solution using a modified U-Net structure and deep learning. Our approach improves image analysis and was tested on two benchmark datasets with high accuracy rates. We also classified tumor sub-regions into three categories. Our results show that our modified U-Net is a powerful tool for brain tumor diagnosis and treatment planning, outperforming existing methods.

Arslan et al. [48] developed a health monitoring system integrating sensors, wireless communication, and machine learning to analyze patient vitals. Using KNN, SVM, and SGD algorithms, they achieved classification accuracies of 92 %, 94 %, and 97 %, respectively. This system significantly enhances healthcare monitoring by ensuring precise and real-time patient health assessments. Qiu et al. [49] highlighted deep learning's role in improving Wireless Sensor Network (WSN) deployment. Their research demonstrated enhanced energy efficiency, speed, and accuracy, enabling better decision-making in healthcare, agriculture, and environmental monitoring. Future work emphasizes privacy-preserving techniques and scalable models to address dynamic environments and unique WSN challenges. Jameil and Al-Raweshidy [50] proposed a cloud-based digital twin framework for real-time healthcare monitoring. Integrating multimodal sensors and XGBoost, they achieved 98 % predictive accuracy with reduced latency by 32 %. Their system advances proactive health interventions while addressing critical constraints like latency, scalability, and predictive accuracy in IoT healthcare applications. Saini et al. [51] reviewed IoT and Wireless Sensor Networks (WSNs) in healthcare, emphasizing predictive analytics, remote monitoring, and telemedicine. These technologies revolutionize patient care but require addressing privacy and security issues. Future directions involve integrating IoT data with health records and AI-driven analytics to enhance healthcare outcomes.

Though most of the earlier researchers have utilized a machine learning model for the WSN healthcare system, it has difficulty processing large data sets. And in some cases, the ML required more time to process the input patient records. They needed more features to establish communication bands between patients, doctors, and healthcare units. To improve the efficiency and capacity of the WSN model, an advanced learning technique is required. So, in this paper, deep learning-based models are implemented to control and manage patient records in the healthcare sector.

4. Problem statement

All medical industries are growing daily due to the increasing number of patients, new diseases, and doctors required to treat them. Medical industry management uses human resources, which are called managers, and software, which is called medical industry management. The managers manually arrange resources for the out-patients and in-patients in all aspects. The software module provides reports, data storage, and information concerning the query feed by the operator. However, medical industries still face continuous problems managing patients, visitors, medical representatives, and patients with the right resources at the right time. In addition, predicting diseases at an earlier stage is one of the major issues in medical industries. Especially for predicting stroke, cancer, heart attack, etc, various methods are traditionally followed, but the accuracy of those methods is not sufficient. It is due to non-standardized input medical data, missing values, imbalanced data, and clinical oversights. This problem is considered the main research problem, and it can be represented in detail here to help you understand and design the solution.

Let $\boldsymbol{N}$ be the number of sections ($\boldsymbol{S}$) in the medical industry/hospital ($\boldsymbol{MI}$), $\boldsymbol{S_i}$ different section in $\boldsymbol{S}$, and $\boldsymbol{K}$ be the total number of patients suggested for each section.

$$\boldsymbol{S} = \{\ \boldsymbol{S_1}, \boldsymbol{S_2}, \boldsymbol{S_3}, \ldots, \boldsymbol{S_i}, \ldots, \boldsymbol{S_N}\} \forall\ \ \boldsymbol{i}\ \ = \mathbf{1} \boldsymbol{toN}$$

$$\boldsymbol{S} \in \boldsymbol{section}(\boldsymbol{MI})$$

Patients ($\boldsymbol{Pa}$) in each $\boldsymbol{S_i}$ is

$$\boldsymbol{Pa} = \{\ \boldsymbol{Pa_1}, \boldsymbol{Pa_2}, \boldsymbol{Pa_3}, \ldots, \boldsymbol{Pa_j}, \ldots, \boldsymbol{Pa_K}\} \forall\ \ \boldsymbol{j}\ \ = \mathbf{1} \boldsymbol{toK}$$

The medical industry $\boldsymbol{MI}$ uses several wireless network sensors ($\boldsymbol{L}$ numbers) to monitor and record various kinds of data predefined to the nodes. Several body wearable sensors called type-1 ($\boldsymbol{t1}$) monitor and record the patient's health data to determine the health condition. Only four sensors called type-9 ($\boldsymbol{t9}$) are attached inside the monitoring section. A few sensors called type-10 ($\boldsymbol{t10}$) are attached to the medical environment to monitor and record the environmental data. Finally, type-11 ($\boldsymbol{t11}$) sensors that monitor the entry and exit locations of the healthcare environment. The $\boldsymbol{t1}$ sensors include several wearable body sensors that can sense data from various parts of the patient's body. Table 1 shows the sensor types and the data that can be sensed from the human body. The sensors ($\boldsymbol{t_i}$) monitor and generate data like the environment, vehicles, and people entering and exiting the MI. Based on the data types, different medical experts and doctors ($\boldsymbol{d_{ij}}$) are assigned at the right time to treat the patient. This paper assumes that $\boldsymbol{K}$ doctors are experts in treating each kind of medical data. The doctor $\boldsymbol{d_{ij}}$ is the $\boldsymbol{j^{th}}$ expert to treat the patients based on the type $\boldsymbol{t_i}$. Table 1 gives the sensor type, sensor use, and doctors assigned for the data analysis. It shows that data preparation and assignment to the concerned doctors are reduced in complexity.

Hospitals these days are teeming with patients with various types of new diseases and with few symptoms. Managing patients has become crucial for the administration in assigning the right medical experts to provide the correct treatment. This problem is continuous nowadays in real-time medical industries, and a better solution is needed to automatically assess the experts and sections for the in-house and out-patients. Hence, this paper proposed a Wireless Sensor Network architecture model for monitoring, analyzing, and assigning patients with sections and appropriate doctors for saving people. A deep convolution neural network model is used to analyze the sensor data for medical industry management and predict the diseases to get the experts.

4.1. Proposed methodology

In this paper, deep learning-based approaches CNN and LSTM are applied with the WSN to accurately monitor and predict various types of diseases from the input medical data. The proposed approach includes data pre-processing, feature extraction, classification, and prediction. Fig. 2 depicts the overall workflow of the proposed model. It starts with data collection, preparation, preprocessing, learning, feature extraction, and prediction. Two deep learning models are used for prediction where: CNN is used for image data, and LSTM is used for continuous data.

5. Data pre-processing

Converting raw data into solid information is known as data pre-processing or data preparation. It mainly consists of four steps: data cleaning, data discretisation, data transformation, and data integration.

6. Data cleaning

The process of changing or removing duplicate, corrupted, incorrect

Table 1
Sensor, data, and doctors.

Sensor Type	Data	Respective Doctors
T1	ECG	d_{11} to d_{1k}
T2	EMG	d_{21} to d_{2k}
T3	Heartbeat	d_{31} to d_{3k}
T4	Pressure	d_{41} to d_{4k}
T5	Chest images	d_{51} to d_{5k}
:	:	:
Ti	Medical data	d_{i1} to d_{ik}
:	:	:
T6	Brain images	d_{61} to d_{6k}
T7	Abdomen images	d_{71} to d_{7k}
T8	Body mass index	d_{81} to d_{8k}
T9	Section environment	d_{91} to d_{9k}
T10	Overall environment	d_{101} to d_{10k}
T11	Entry-Exit	d_{111} to d_{11k}

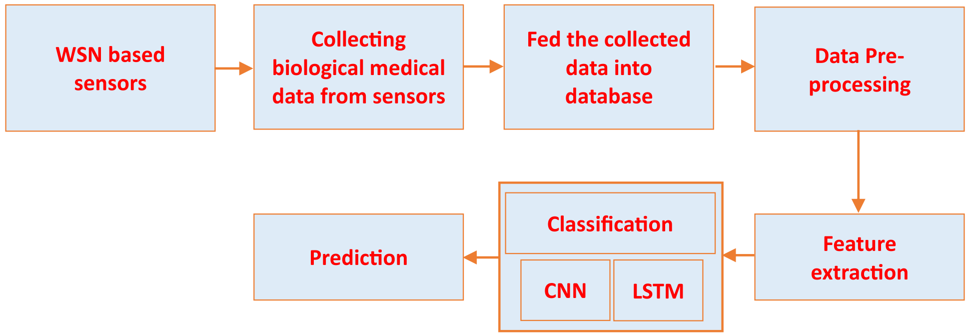


Fig. 2. Proposed work flow.

value, incorrect format and incomplete data into the complete data within the dataset is known as data cleaning. When combining more datasets, there have been more chances for non-labelled or duplicate data. If the data is incorrect, then the algorithms' output is also irrelevant or unsuitable for the input. Correcting or removing them is the process of data cleaning. It has five steps to cleaning the data they are the first step is to remove irrelevant data or duplicate data; the second step is to fix the structural errors; the third step is to filter the unnecessary data or unwanted data; the fourth step is to handle the missing data in the dataset, and the final step is to recheck or validate the dataset.

7. Noise removal

Image denoising is also the process of noise removal from the image. Because some small noise can cause the loss of information. The extra or unwanted pixel values are added to the image while capturing, which causes the loss of information while analysing them. This noise should be classified into various types they are: Impulse Noise (IN), random valued impulse noise (RVIN), salt-and-pepper impulse noise (SPIN) and Additive White Gaussian Noise (AWGN). This noise can occur in many ways, such as damage to the circuit because of the heat, taking images in low-light situations, and the bit error occurring in data transmission for long distances. These are some ways that the noise should occur in the image while capturing. Removing this unwanted or extra noise in the image is known as noise removal.

8. Handling missing values

Real data contains many missing values, which can affect the performance of the whole data. So, it is also one of the crucial methods in the data pre-processing model. Handling missing values or missing data in the whole dataset is known as handling missing values. The handling method should depend on the factors. Before handling the missing values, we first need to track the missing values or data.

9. Repeated values removal

Repeated value removal is the process of eliminating duplicate values or records from the dataset. This process plays a crucial role in ensuring data quality, improving accuracy, and preventing the bias model from overfitting. It improves data consistency, reduces noise, and prevents the model from overfitting.

10. Medical data feature extraction

The process of transforming the raw material data into a group of relevant and meaningful features that can be very useful in various applications, which include disease analysis, risk prediction and planning for treatment. Raw medical data includes electronic health records (EHRs) and hospital patient records. The extracted feature is used to develop a machine-learning model to identify the outcomes of the patients. By analysing the relationship and the pattern from the medical data, this feature is used to analyse the patient's disease and their stage. By analysing the current stage and the patient's medical history, this feature is used to make the treatment plans for the patient. It provides several advantages to this medical industry: it improves data accessibility, it helps better in the decision-making process, and it also provides the data integration process.

10.1. Proposed WSN model

The proposed WSN indicates the set of sensor nodes deployed in the healthcare industry in such a manner to monitor and record various data about sections, patients, and other related concerns. All the sensors generate their own defined data based on their nature and behavior. The wireless sensors' data includes time series data, signals, images, and video. Based on the data generation, the related sensors are deployed correctly in the healthcare industry. For example, signal-sensing sensors are placed on the patient's body, like around the heart, wrists, and knee, to generate ECG data. Similarly, different kinds of sensors are installed at different locations inside the healthcare industry and patients. The sensors installed in the healthcare industry are illustrated in Fig. 3. The network involves certain components, like

sections as $S = \{S1, S2, \ldots Si, \ldots Sn\}$

patients with sensors $P = \{P1, P2, \ldots, Pi, \ldots, Pm\}$

and the sensor hubs (H) collect and aggregate the data, which is

$H = \{H1, H2, \ldots, Hi, \ldots, Hs\}$

In addition to the above components, it is considered that the patients deployed with the sensors are in ICU. All the hubs collect and aggregate the data and forward it directly to the deep learning framework, where multiple deep learning algorithms are used to analyze different kinds of data obtained from the sensors. For example, the image data is processed using the CNN algorithm, time series data are analyzed using the LSTM algorithm, and so on. Fig. 3 depicts the overall workflow of the proposed WSN model on predicting various health records from the patients and assigning doctors, treatment, and sections in the medical sector in detail.

10.2. Deep learning for processing wireless network sensor data

Many deep-learning algorithms can process data generated from different sensor nodes in the WSN-based medical environment. This paper used two different deep learning algorithms, CNN and LSTM, for processing images and other time-series data, which are explained below.

10.3. Convolution neural network

Recently, Artificial Intelligence (AI) technologies have been updated

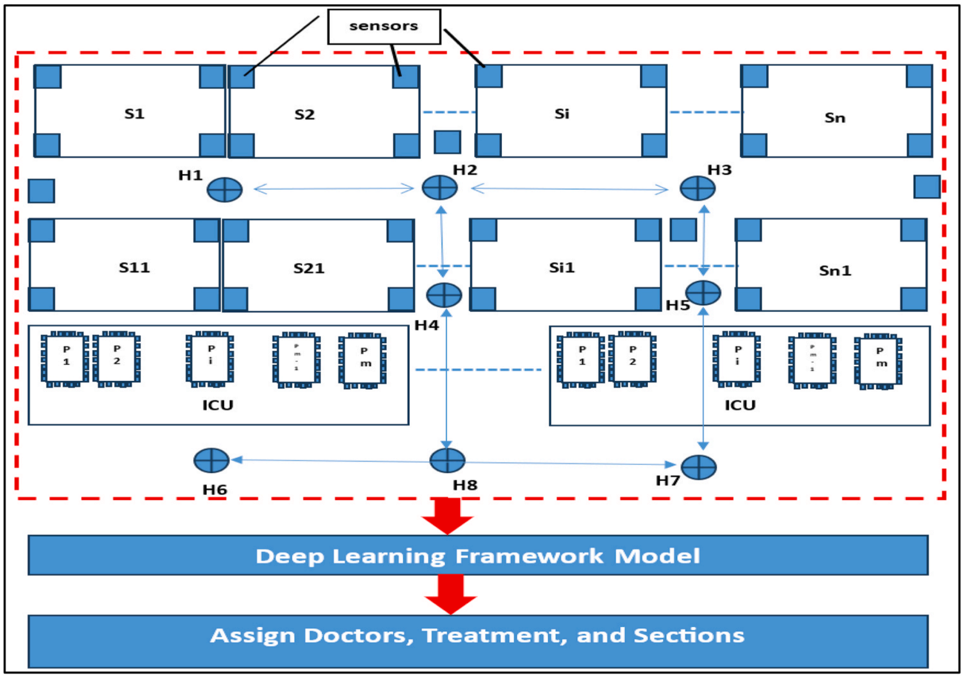


Fig. 3. Proposed wireless sensor network model.

occasionally. Various apps are being developed daily, particularly in image transformation, identification, etc. Some of the Image-transforming apps are DeepArt.io and Prisma, which transform photos into artworks using different styles of well-known artwork. There are so many advancements in various fields, such as diagnosing various diseases by a high precision reading of medical images like CT and MRI scans, which require minimum involvement of invasive methodology. This helps in the early diagnosis of many acute diseases. In the field of automobiles also, AI is gradually increasing its foot marks, and autonomous vehicles are controlled by AI by various signals and appropriate learning methods. The overall workflow of the proposed model is shown in Fig. 4.

The above-mentioned advancements in AI are possible through learning methodologies, particularly deep learning. AI is a technique that mimics the intelligence of humans through computers with certain rules, logic, and machine learning. Machine Learning has algorithms to enhance decision-making by reading the data. Likewise, the upcoming requirements for solving complex problems demand machine learning with additional algorithms, allowing the software to train itself to execute various tasks. It is achieved with the help of deep neural networks to process huge amounts of data. The flow of data generation, transmission, and processing in the healthcare industry is shown in Fig. 5. The sudden rise of massive data in today's digital world, which continues to grow further, has brought many changes like decreased storage cost, more image and text data transactions, etc. Moreover, the vast GPU availability has achieved less cost, high power, and fast processing.

10.4. Convolutional neural network

Among the various classes of Neural Networks, the Convolutional Neural Network gives promising results in Natural Language Processing, Computer Vision, and Speech recognition. Specifically, it has remarkable results in image recognition and classification with various applications like power vision, face recognition, etc., in self-driving cars and robots. The convolutional neural network contains three major layers: the convolutional layer, the pooling layer, and the fully connected layer, as shown in Fig. 6, which shows the arrangement of layers with different

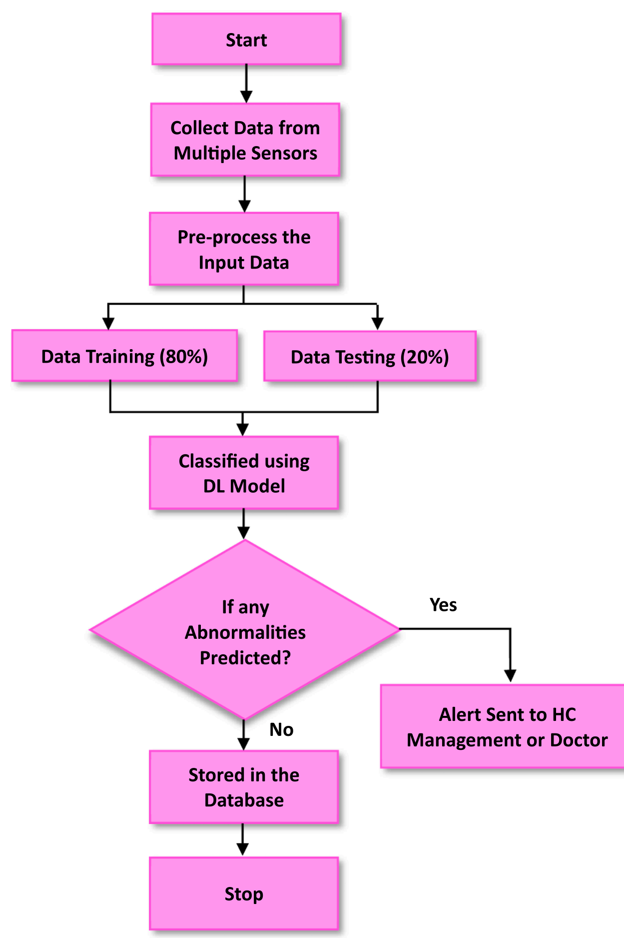


Fig. 4. Proposed model flowchart.

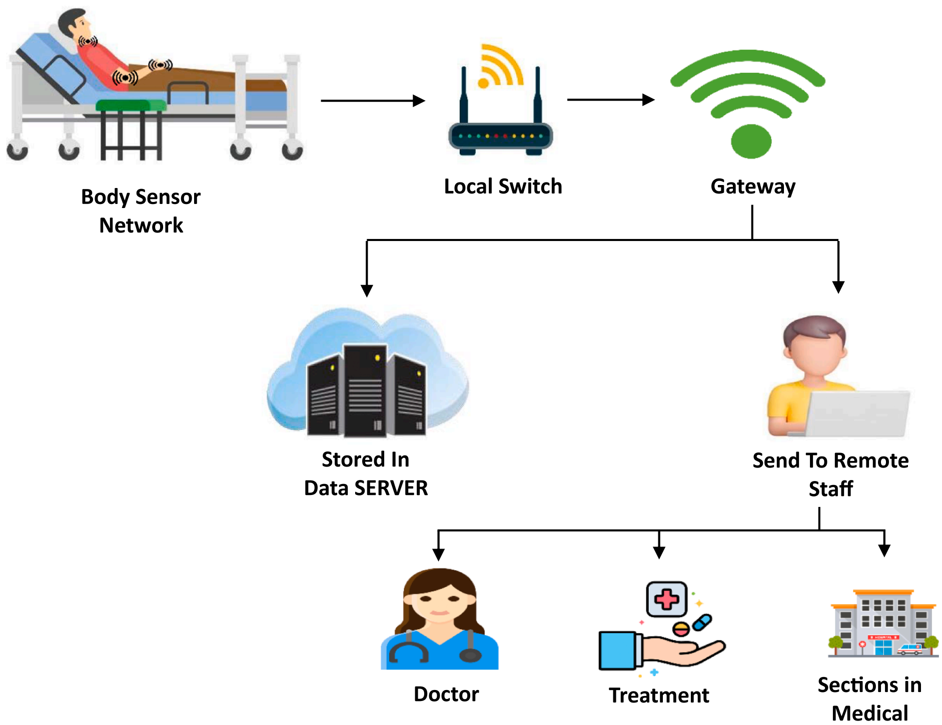


Fig. 5. Data processing flow.

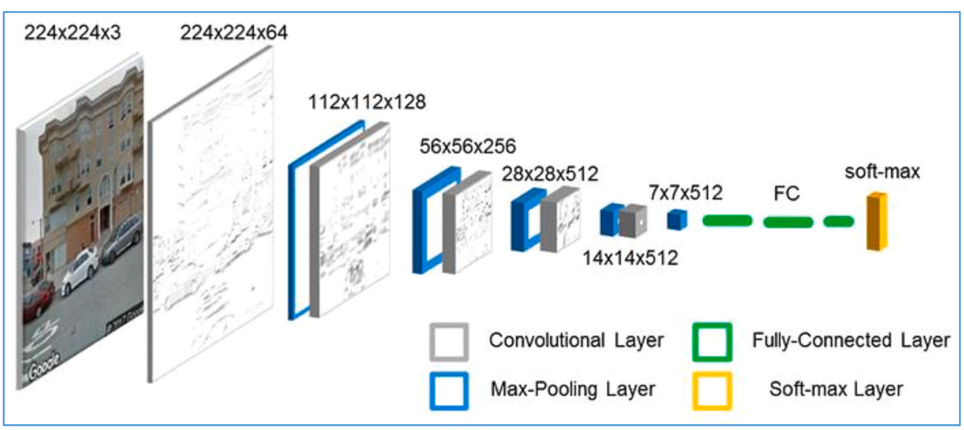


Fig. 6. Convolutional neural network architecture with different layers.

color shades. Gray, Blue, Green, and Yellow represent the Convolutional Layer, Max-Pooling Layer (Pooling Layer), fully connected, and Soft-Max layers. In the above figure, an image is classified based on the combination of red, green, and blue (RGB) colors in the camera's captured image. This color code is then transferred into grayscales of 2D matrix format and given to the CNN, where the pixel values of the matrix range from 0 to 255.

10.5. Convolutional layer

The convolution layer is a major building block of CNN's architecture [52]. As a result, CNN is named after the convolution layer, which has a convolution operator. The main aim of the convolution is to observe the Input Image's feature. So, it acts as the "feature detector" of the CNN algorithm. The matrix slides over the original picture by 1 pixel, and a sum of elementwise products in all positions gives the output matrix with a single element. The matrix that slides is termed a "Filter," and the resulting output Matrix after the filter sliding is termed a "Feature Map" or "Activation Map." Different Filters must be applied to the same image to obtain various other features or activation maps. The matrix values need to be altered to create different Filters, where different filters will identify different features. During the training, CNN will learn the values of the filters by themselves. With more filters, more features can be obtained from the image, and by repeating the training, the CNN will get well-trained to identify the patterns in the unseen images. In a CNN layer, the operation between the input image and the mapped features is defined as:

$$\boldsymbol{Y}_{ij}^{K} = \sum_{m=0}^{M-1}\sum_{n=0}^{N-1}\boldsymbol{X}(\boldsymbol{i}+\boldsymbol{m})(\boldsymbol{j}+\boldsymbol{n}) \bullet \quad \boldsymbol{W}_{mn}^{k} + \quad \boldsymbol{b}^{k}$$

The Kth kernel's mapped features at position (i, j) are represented by Y_{ij}^{K}, which is calculated by convolving the input image X with the weight of the Kth kernel, W_{mn}^{k}, and adding the bias value, b^{k}. In the convolutional layer, an activation function introduces non-linearity, producing negative and positive values, or 0 and 1, respectively. It is defined as:

$$A_{ij}^{k} = \textbf{ReLU}\left(Z_{p,q}^{k}\right) = \boldsymbol{max}(\mathbf{0}, Z_{p,q}^{k})$$

10.6. Pooling layer

The second important layer in CNN is the “Pooling Layer.” The Pooling layers are next to the Convolutional Layers; in other words, they are present between Convolutional Layers. This layer generally reduces the dimensionality of the activation or feature maps while retaining the key information derived from the input image. This process of transforming Information from Input is called “downsampling.” The various methods of Pooling are Max Pooling, Average Pooling, and so on. The dimensionality of the input-mapped images is reduced by expressing:

$$P_{ij}^{k} = \textbf{max}\left(A_{p,q}^{k}\right), \quad \boldsymbol{p} \in [\boldsymbol{i}, \quad \boldsymbol{i}+\boldsymbol{s}], \quad [\boldsymbol{j}, \quad \boldsymbol{j}+\boldsymbol{s}]$$

Here, P_{ij}^{k} represent the pooling layer output generated from $\mathbf{K^{th}}$ feature map at **i, j** position, $\textbf{max}\left(A_{p,q}^{k}\right)$ indicates the maximum activation function value obtained from the previous layer and s indicates the size of the pooling.

In Max Pooling, a 2×2 Window size is defined, and it takes the maximum value of the element in the feature map of the same window. Similarly, the average value of the window is calculated in the average pooling. Comparing both pooling layers, the max pooling stands out in many cases and reduces the feature map’s dimensionality better than the average pooling layer. The number of feature maps in pooling layers is the same as that developed by the convolutional layer when the pooling operation is applied. Only the dimensions are reduced after the pooling operation is applied. This Pooling operation is also called Downsampling. The major advantages of Downsampling are that it is smaller, computation is easy due to the limited number of parameters, Overfitting controls any deviations or changes in input image transformations, and it reduces the data representations.

10.7. Fully connected layer

The fully connected layer is the final layer, which gives the network’s output by calculating the class scores. The output volume’s size is $1 \times 1 \times$ N, and the value of the “N” is the total output class number. This layer’s neurons are connected to every neuron present in the layer and the previous layer’s neurons. In a few CNN architectures, the number of fully connected layers is increased in the end part. Due to this, the Convolutional and Pooling Layers will observe more features in the input image. The fully connected layer utilizes the greater number of features extracted to categorize the image’s different classes depending on the dataset used in training.

$$F_i = \sum_{j=0}^{N} A_j \bullet W_{ji} + b_i$$

The final output of the model on the ith class, denoted as F_i, is calculated using the activation result from the previous layer, A_j the weight value between the jth to ith class in the model, W_{ji}, and the bias value of the ith class, b_i. These components work together to generate the final output, F_i, which represents the model’s prediction for the ith class.

10.8. Streaming layers

The streaming layers have a stream of services to the data sources, also called feature layers. The original datasets include live observations, abnormal people entry, location changes, and other attribute changes. These stream layers have features like polylines, points, or polygons, and unlike feature layers, they provide their data source, and they will call on the data to listen to the active stream of data. Most of the time, irregular intervals are broadcast, and the researchers use the Spark streaming layer for data streaming to diagnose. The data is received from distinct sources to obtain high-speed streaming and transformed into minimum batches.

10.9. Long short-term memory

LSTM displays the performances when large datasets are handled, and these datasets are used to increase memory usage with computational complexity. The proposed models utilize the firefly optimization approach with the LSTM model, which gives hyperparameters to reduce drawbacks. These hyperparameters are epochs and hidden layers needed for better optimization than the LSTM model. To ensure the global best function, the prediction is performed the diagnosis to the highest rate. The fitness function is mathematically expressed in the equation below for finding the *Gbest* value. They select the random hyperparameters and pass them to the LSTM training. For every iteration, parameters are calculated, and the fitness function is matched when the iteration is stopped. The deep co-learning model analyzes the labels for phrases, and other phrases can’t be defective to the classifier. The Architecture Long Short-Term Memory (LSTM) model is shown in Fig. 7.

The output of the LSTM cell is denoted as $\boldsymbol{ht}$, $\boldsymbol{t}$ is the memory cell value, and the LSTM cell outputs two values, as shown in Fig. 6. Long short-term memory (LSTM) model architecture is represented as $\boldsymbol{ht}-\mathbf{1}$. The input data of the LSTM cell is defined as $\boldsymbol{xt}$ operating at the time $\boldsymbol{t}$. The process of calculating the LSTM unit is explained in the following steps:

The weight matrix is represented as W_c which is as shown in Eq. (1). $\widetilde{c}_t$ is known as the candidate memory, which is calculated, and the bias is represented as b_c.

$$\widetilde{c}_t = \textbf{tanh}(W_c.[h_{t-1}, x_t] + b_c) \tag{5}$$

The input gate (i_t) is the current input data that updates the memory cell’s state value and controls the input gate. The bias is represented as b_i and the weight matrix is represented as W_i, The sigmoid function is denoted as $\boldsymbol{\sigma}$, shown in Eq. (2).

$$i_t = \sigma(W_i .[h_{t-1}, x_t] + b_i) \tag{6}$$

f_t *is* the forget gate calculates the memory state value obtained based on the historical data that updates and controls the forget gate. The bias is represented as b_f and the weight matrix is represented as W_f, as given in Eq. (3).

$$f_t = \sigma(W_i .[h_{t-1}, x_t] + b_f) \tag{7}$$

The current moment memory cell c_t is evaluated, and the value for the last LSTM unit is denoted as c_{t-1}, as given in Eq. (4).

$$c_t = f_t * c_{t-1} + i_t * \widetilde{c}_t \tag{8}$$

‘* ’ denotes the dot product. Forget gate controls update the memory cell based on the state value for the last cell candidate value.

$$o_t = \sigma(W_o .[h_{t-1}, x_t] + b_o) \tag{9}$$

Where, o_t is known as the output gate; this calculates the memory cell state value as the output gate controls the output. The bias b_o and the weight matrix is denoted as W_o. The output h_t for the LSTM cell is calculated as shown in Eq. (6).

$$h_t = o_t * \textbf{tanh}(c_t) \tag{10}$$

Based on memory cells and control gates, the LSTM model easily updates, resets, reads, and keeps long-term information. It also has a sharing mechanism of internal parameters that control the output dimensions based on weight matrix dimensions settings.

LSTM is needed for continuous data processing because it can eliminate the same data with the same operations multiple times using the forget gate. Also, it avoids setting more epochs, which increases the computational complexity by comparing the present data processed output and compare with the previous output, which can eliminate memory storage and avoid further processes on the same data and its output. The above-discussed CNN and LSTM models are implemented in Python to analyze different healthcare industry data generated by many wireless sensors deployed in the environment. The wireless sensor network creation is simulated in NS2 software, and the traffic flow is understood. Based on that network flow, the sensor data are collected minimum and used in the experiment. To verify the performance of the proposed CNN and LSTM models, various healthcare datasets are collected from benchmark resources that are freely available on the Internet, like Kaggle and GitHub. The entire process of the paper is given as a step-wise procedure that helps to do programming and implementation with fewer errors that can be rectified easily and quickly to improve the accuracy of disease prediction. The entire process of this paper is given as a stepwise procedure in the name of Deep_Learning_Model. Initially, all the data that comes from the sensors are verified as numeric, alphanumeric, video, or images. Based on the data types, they forward it to the corresponding deep learning model as LSTM or CNN. Then, 80 % of the respective data is used for training the models, and 20 % of the remaining data is used for testing (10 %) and validating (10 %) the models. The classified output from the deep learning model is combined to forecast it to the admin.

Deep_Learning_Model (data array [])

```
{
        Input data;
        Check the datatype of data;
        If (data.type == numeric) || (data.type == alpha-numeric)
                Send to LSTM( )
        Else
        If (data.type == image) ||(data.type == video)
                Send to CNN( )
        Train-data = 80% (data)
        Test-data = 20% (data)
        O1= LSTM.train(Train-data.numeric)
        O1= LSTM.test(Test-data.numeric)
        O2=CNN.train(Train-data.images)
        O4=CNN.test(Test-data.images)
        Output = Combine (O1, O2, O3, O4);
}
```

The above procedure can be implemented in any programming language to verify the performance of the proposed deep-learning models.

10.10. Experimental setup

The proposed algorithms and the procedure followed in this paper are simulated and implemented in NS2 and Python, respectively. The simulation of WSN for the healthcare industry is carried out in NS2, and the data analytic models are implemented in Python. This software is installed in Intel Pentium Core dual processor i5, 11th generation, with 1TB HDD and 16 GB RAM. The system configuration is verified before

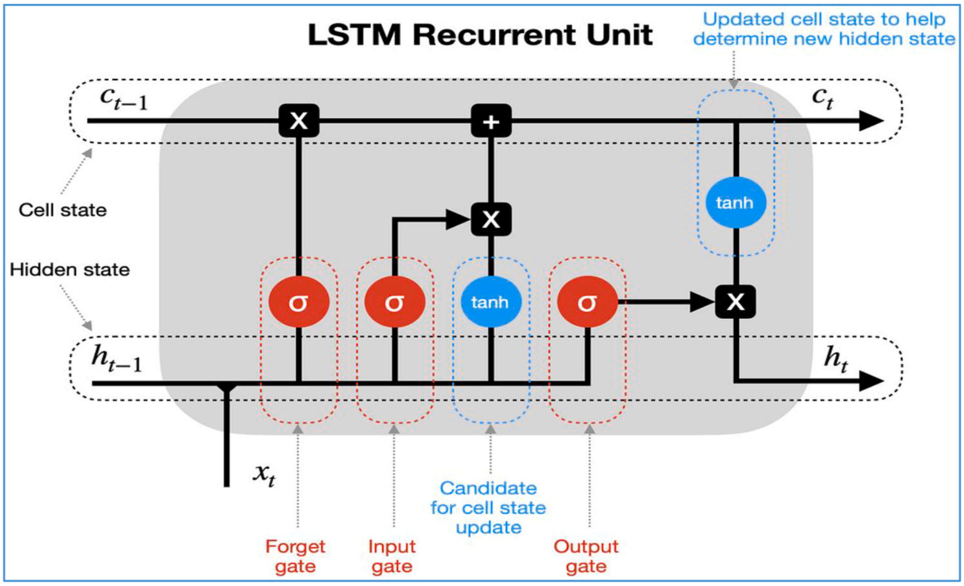


Fig. 7. Long short-term memory (LSTM) model architecture.

executing the programs to avoid unnecessary exits. The network architecture of the proposed model is simulated in NS2 software. The IoT devices are deployed as WSN nodes, and their communications are created as links using UDP and TCP/IP. The data size and the time interval of pocket transmission are well defined in the NS2 code, along with various network parameters like area, number of nodes, packet size, propagation model, and antenna. The simulation is conducted for 20 seconds, verifying their network performance. To validate the deep learning models' performance, the traffic data from the routing table generated in NS2 is used with a benchmark dataset in Kaggle. The experiment uses input data samples taken from the publicly available stroke prediction dataset in Kaggle. It includes 5110 data under 12 different attributes. The input dataset contains unique patient ID, gender, age, marital status, work type, residence type, BMI level, etc. This research mainly focuses on predicting whether patients are affected by stroke using various parameters. The dataset is split into three portions for training, testing, and validation. Initially, 80 % of the whole data, called training data, is used for training the models. The remaining 10 % is used for validation and 10 % for testing. The layer configuration of the models obtained during the training process is shown in Fig. 8a) and b) for CNN and LSTM, respectively.

This section discusses various results obtained from the proposed DL model on predicting the presence of stroke from the patient's medical data. Fig. 9 depicts the age distribution ratio among male and female patients. The X-axis represents the age, and the Y-axis represents the count. The analysis result shows that most of the patients in the age group of 60–82 have widely affected by stroke. Some of the female patients from the age of 15–60 have also been affected by stroke. Fig. 10 depicts the Gender-wise distribution of stroke patients. The X-axis represents the gender, and the Y-axis represents the age. The analysis result shows that most of the male patients in the age group of 50–60 are widely affected by stroke. And female patients from the age of 20–55 have been affected by stroke. The result depicts that compared to men, women are affected by stroke at an early age.

Fig. 11 (a) depicts the gender-wise ratio of stroke patients analyzed by the proposed model. The analysis result shows, from the total number of input data, 42.58 % are male patients, and 57.42 % are female patients who had a stroke. Fig. 11 (b) depicts the total number of male and female stroke patients. The analysis result indicates that 120 female and 89 male patients are predicted as having a stroke.

Fig. 12 (a) depicts the smoking status of the stroke patients based on the input parameter. To find the smoking status, the analysis result is classified into four categories: never smoked, formerly smoked, smokes, and unknown. The analysis indicates that from the overall dataset, 40.19 %, 13.88 %, 18.66 %, and 27.27 % are predicted as never smoked, unknown, smokes, and formerly smoked patients, respectively. Fig. 12 (b) depicts the total number of patients having smoking behavior. Based on the smoking type, it is classified into four categories. The result shows that in this input sample, 84 patients are never smoked, 57 are formerly smoked, 39 are smokers, and 29 are unknown.

Fig. 13 (a) depicts the gender-wise marital status of the stroke patient. Based on the analysis records, the patient's marital status is predicted. The result shows in the input sample that 89.00 % of the patients are married, and the remaining 11.00 % are unmarried. Fig. 13 (b) shows the marital status count of the stroke patient analyzed by the proposed approach. The analysis result emphasizes that most stroke patients are married, with a count of 186, followed by the unmarried patient's count of 23. Fig. 14 (a) illustrates the status of the BMI distribution among male and female stroke patients. The X-axis indicates the BMI distribution ratio, and the Y-axis indicates the count. The analysis result shows both male and female stroke patients' BMI falls between 25 and 35. In some cases, the BMI value increases too high. Fig. 14 (b) represents the Gender-wise BMI distribution among stroke patients. The analysis showed that most of the male patients' BMI levels fell between 25 and 35, and the BMI level of female patients fell between 23 and 31.

Fig. 15 (a) shows the number of stroke patients with heart disease. Fig. 15 (b) shows the presence of heart disease among stroke patients. The prediction result indicates that 80.86 % of the patients are not affected by heart disease, and the remaining 19.14 % have heart disease. That is, only 40 patients have heart disease, and the remaining 169 patients do not have any heart disease.

Fig. 16 (a) shows the status of hypertension among stroke patients. The prediction result indicates that 71.29 % of patients do not have hypertension, and the remaining 28.71 % have hypertension. Fig. 16 (b) shows the number of stroke patients having hypertension. That is, only 60 patients have hypertension, and the remaining 149 patients do not have hypertension.

Fig. 17 shows the average glucose level distribution among male and female stroke patients. The prediction result is graphically represented in Fig. 17(a). It is clear from the figure that the glucose level falls between 60 and 120. The graphical result in Fig. 17 (b) shows that most male and female patients' glucose levels fall between 70 and 120 and 55–115, respectively. Fig. 18 (a) represents the occupation type of the stroke patients. Generally, the occupation type is classified into three categories: private, self, and government. The analysis result illustrates that most stroke patients are occupied at private jobs with a ratio of 61.06 %, 25.48 % of patients are self-employed, and the remaining 13.46 % work in government jobs. Fig. 18 (b) illustrates the count of patients working in different sectors. The analysis shows that 127 and 28 patients work in the private and government sectors. 28 patients are self-employed, and the remaining 1 are children.

Fig. 19 (a) represents the residence type of stroke patients. The prediction result of the proposed approach indicates that 52.15 % of the patients are from urban areas, and 47.85 % are from rural areas. The count of patients in both rural and urban areas illustrated in Fig. 19 (b) depicts that around 109 patients live in urban areas, and 100 patients live in rural areas. Fig. 20 shows the correlation map result of the proposed model for detecting diseases using various parameters. Based on the actual and predicted correlation results, the positive and negative values of the diseases are identified. The result of the analysis showed that there is no high correlation between the features, but a small positive correlation was predicted between age and other features. Similarly, a small positive correlation is predicted between age and hypertension, stroke, average glucose level, heart diseases, and BMI. Between smoking and marital status, occupation type, and BMI, a small positive correlation is predicted. A medium positive correlation is detected between age and occupation type. And between age and marital status, a medium negative correlation is predicted.

Once all the features of input data are analyzed, the overall samples in the dataset are classified into two phases: training and testing. For training, 80 % of the data are used, and for testing, 20 % of the data are used. From the input data, 3926 samples are used for training, and 982 samples are used for testing, as shown in Fig. 21. Now, the accuracy and loss rate of the model are predicted using the following equations: (1) and (2). The accuracy rate indicates the model's overall performance in predicting the total number of positive and negative data in the input. The loss rate defines the error that occurred during data classification.

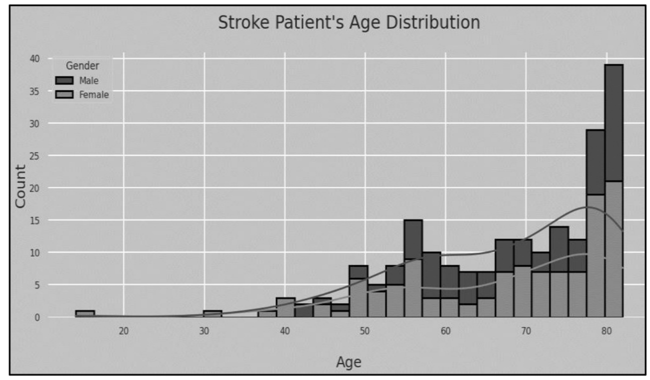


Fig. 9. Age-wise distribution of stroke patient.

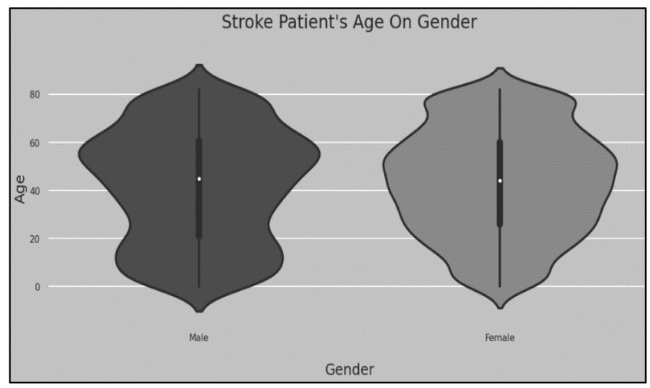


Fig. 10. Gender-wise distribution of stroke patients.

$$Accuracy = \frac{Number\ of\ positvely\ predicited}{Total\ Number\ of\ input\ smaples} \tag{1}$$

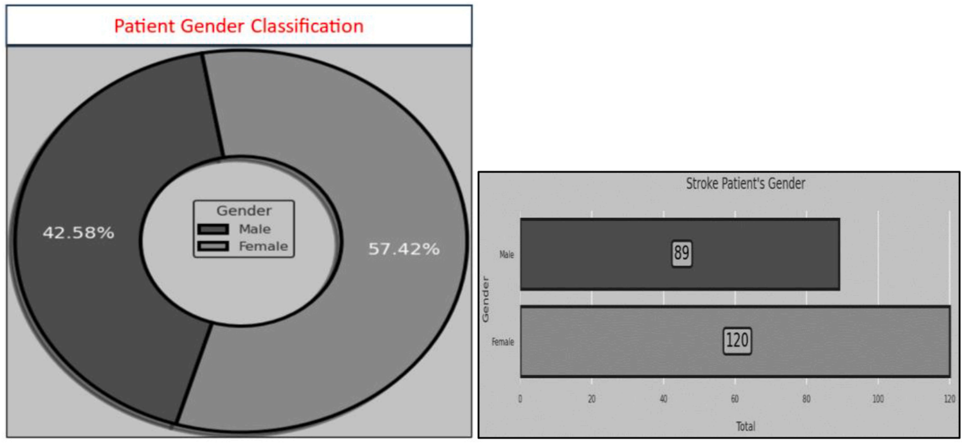
```
Model: "sequential_2"
_________________________________________________________________
Layer (type)                 Output Shape              Param #
=================================================================
dense_10 (Dense)             (None, 32)                352
_________________________________________________________________
dense_11 (Dense)             (None, 64)                2112
_________________________________________________________________
dense_12 (Dense)             (None, 128)               8320
_________________________________________________________________
dropout_2 (Dropout)          (None, 128)               0
_________________________________________________________________
dense_13 (Dense)             (None, 16)                2064
_________________________________________________________________
dense_14 (Dense)             (None, 1)                 17
=================================================================
Total params: 12,865
Trainable params: 12,865
Non-trainable params: 0
```

(a). CNN

```
Model: "sequential"
_________________________________________________________________
Layer (type)                 Output Shape              Param #
=================================================================
lstm (LSTM)                  (None, 10)                480

dense (Dense)                (None, 10)                110

dense_1 (Dense)              (None, 5)                 55

dense_2 (Dense)              (None, 2)                 12

=================================================================
Total params: 657
Trainable params: 657
Non-trainable params: 0
```

(b). LSTM

Fig. 8. Layer architecture.



(a). **Gender-Wise Classification** (b). **Gender Counts**

Fig. 11. Analysis on gender.

$$LogLoss = \frac{-1}{N}\sum_{i=1}^{N}\sum_{j=1}^{M} y_{ij} * \log(p_{ij})$$

Here, in equation (2), N and M represent the total number of rows and columns, y_{ij} indicates the input sample (i) in the j^{th}class, and p_{ij} represents the probability of sample (i) in the j^{th}class, respectively. Fig. 22 shows the accuracy and loss rate of the proposed deep learning-based model for predicting brain stroke. The analysis shows the proposed model achieved 96 % training accuracy with a below 0.30 loss rate. (Fig. 22)

After finding the model's accuracy and loss rate, its overall performance is evaluated using various performance metrics, such as precision, recall, F1-score, and accuracy. These metrics are evaluated using the following Eqs. (11–14).

$$Precision = \frac{TP}{TP + FP} \tag{11}$$

$$Recall = \frac{TP}{TP + FN} \tag{12}$$

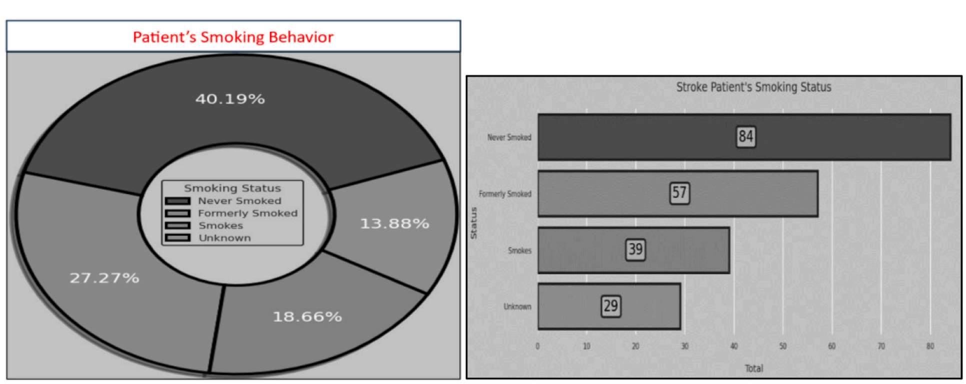
$$F1 - Score = 2 * \frac{Precision * Recall}{Precision + Recall} \tag{13}$$

$$Accurcay = \frac{TP + FP}{TP + TN + FP + FN} \tag{14}$$

Fig. 23 shows the predicted performance metrics results of the proposed neural network models LR, Gaussian naïve Bayes, Bernoulli naïve Bayes, support vector machine, Random Forest, K nearest neighbour, and Extreme Gradient Boosting. The analysis result of Fig. 23 shows that the LR, Gaussian naïve Bayes, Bernoulli naïve Bayes, support vector machine, Random Forest, K-nearest neighbour, and Extreme Gradient Boosting models have achieved 94.6 %, 86.05 %, 93.58 %, 94.6 %, 94.6 %, 94.5 %, and 93.79 % of accuracy, respectively. The overall performance metrics result indicates that the LR, SVM, and RF models have performed better with a high accuracy of 94.6 %.

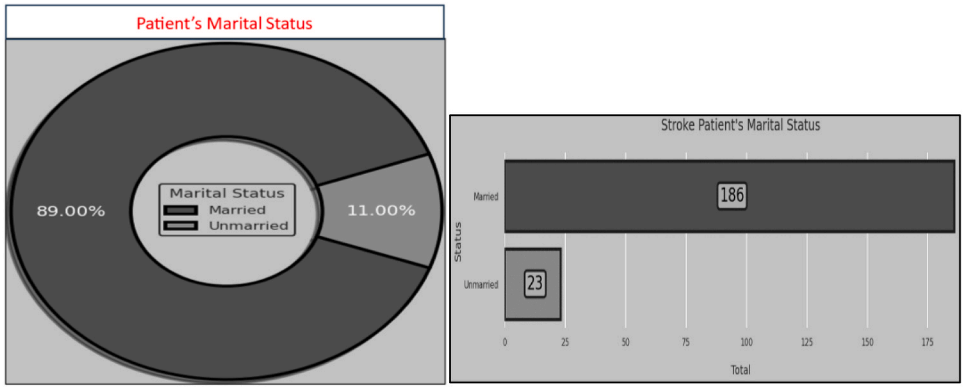
Fig. 24 shows the performance comparison result of the proposed and existing models FRNN [24], DBSCAN [25], and NB [26]. The analysis shows that the FRNN, DBSCAN, NB, and proposed have achieved 90.8 %, 78.83 %, 94 %, and 95 % accuracy. The overall analysis shows that compared to other systems, the WSN is more suitable for many real-time applications, especially for the healthcare sector. Fig. 25 shows the accuracy and loss comparison obtained from CNN during the training and validation. CNN is used to analyze the breast mammogram images to predict the breast cancer. CNN experimented on the breast mammogram images and obtained 98.25 % accuracy and 0.0597 % loss. CNN provides a high rate of true positives in predicting cancer from breast mammogram images. Thus, CNN and LSTM perform individually based on the data types. In the case of multi-modality datasets, multiple deep learning algorithms can process the healthcare data simultaneously to save the patients by providing appropriate treatment at the early stages.

From the above discussion and the results, it is found that the WSN is



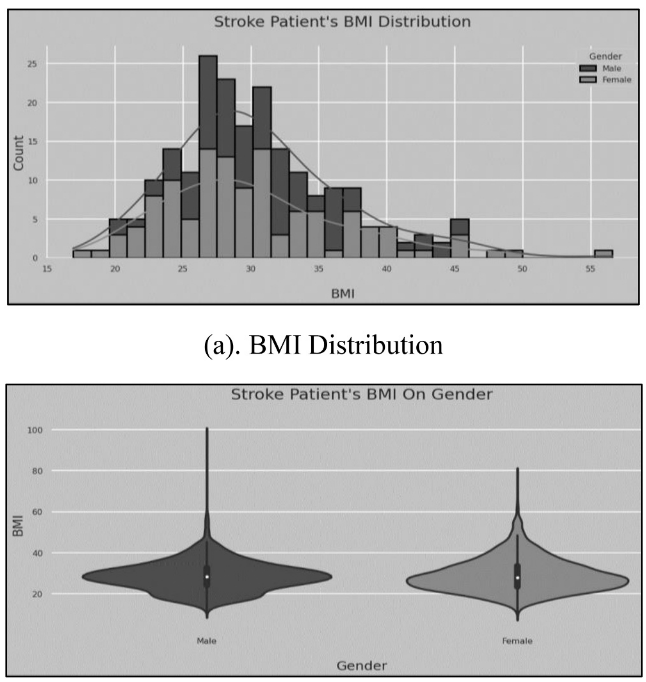
(a). Smoking Behaviour (b). Smoking Patient's Count

Fig. 12. Analysis of smoking behaviour.



(a). **Gender: Marital Status** (b) **Patient's Count**

Fig. 13. Analysis of marital status.



(a). BMI Distribution

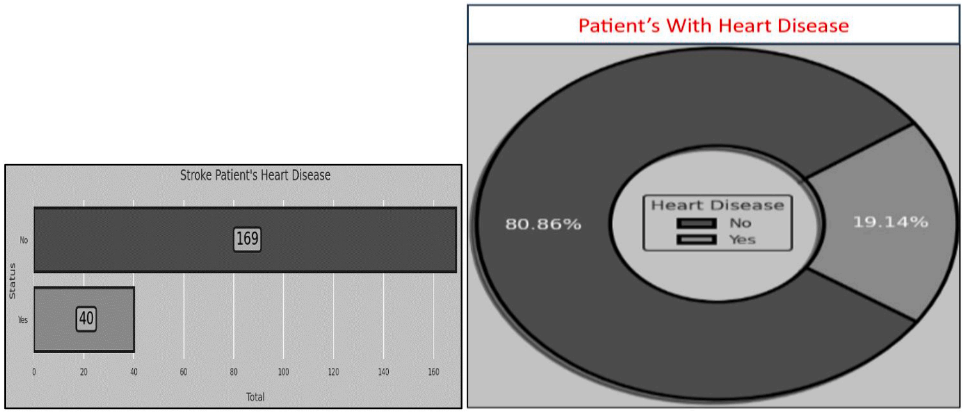
(b) BMI _ Gender Based

Fig. 14. Analysis of BMI.

highly useful in monitoring and controlling the entire healthcare industry remotely, and the data can be analyzed using deep learning algorithms to make predictions easily and speedily. It helps to take immediate action to save the people. This paper provides only a small amount of output, where the output generated by the deep learning algorithms is high in real-time due to various types of sensors generating huge volumes of data.

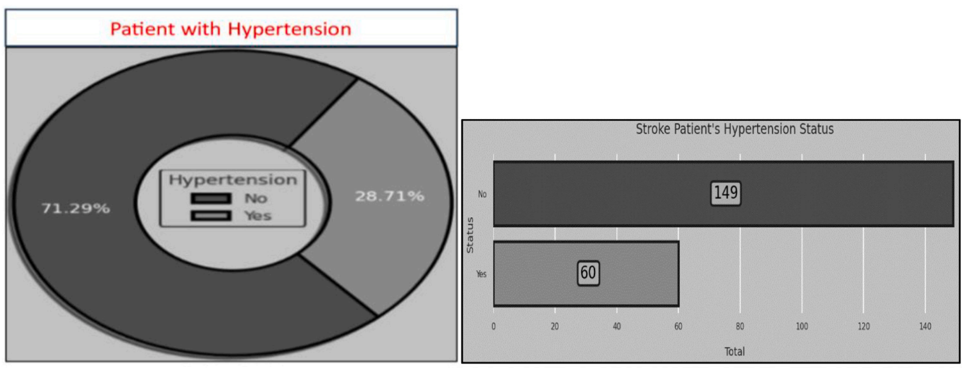
10.11. Limitation of WSN in medical

The major challenge of WSN is security, which means that sensitive information like personal health records should be personal and secure when data is transmitted over WSN. However, security methods, including authentication and encryption, cannot extend adequate security for WSNs. Traditional technologies, like IoT with WSN, have developed with new challenges. Thus, data security must be maintained effectively. Privacy should be trusted and satisfied by the customer, and personal security and sensitive data are the customer's primary concerns. In recent days, WSN has faced low energy constraints in batteries for sending and processing data; WSN needs to overcome this challenge by balancing energy constraints. WSNs face a challenge in balancing energy constraints with the need for high power to process and transmit



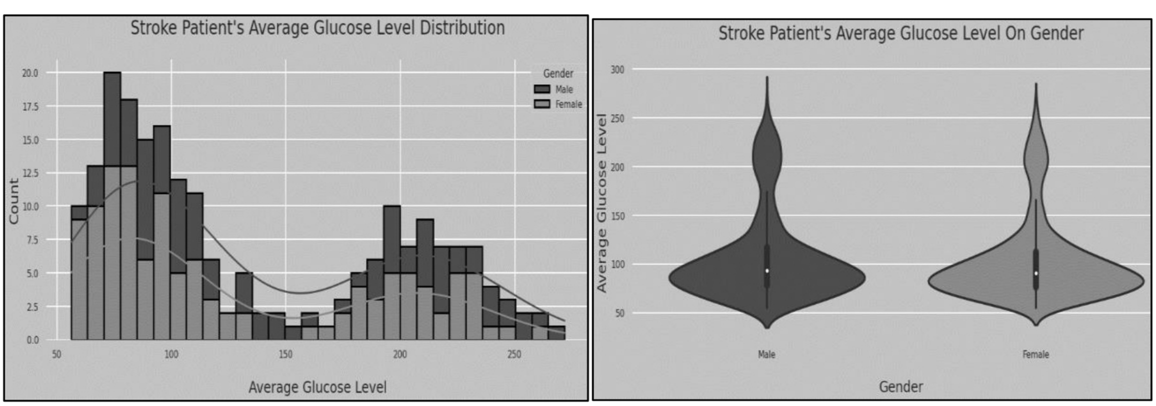
(a). Patients Count (b). Ratio of Patients' Affected by Heart Disease

Fig. 15. Analysis of heart disease.



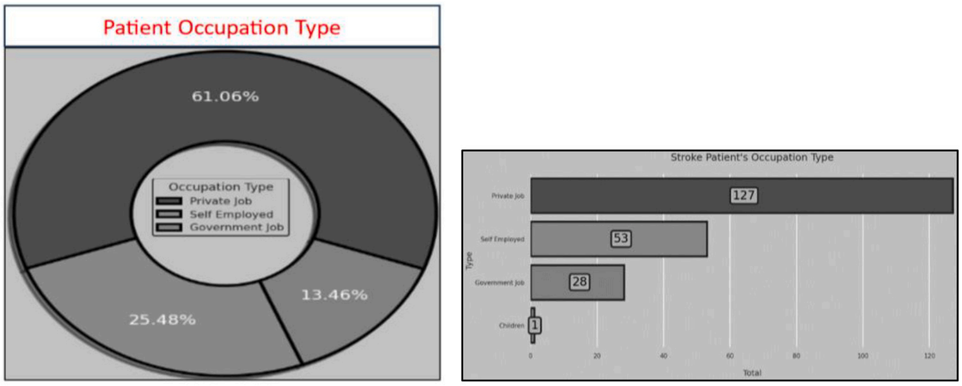
(a). Ratio of Patients Having Hypertension (b). Count

Fig. 16. Analysis of hypertension.



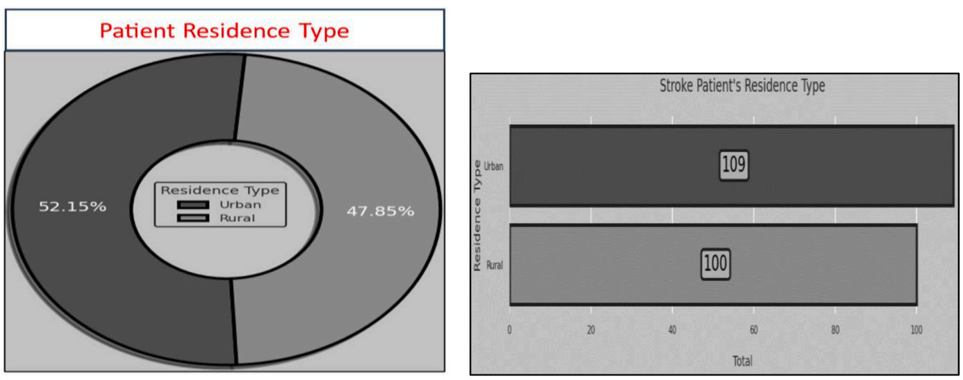
(a). Glucose Level Distribution (b) Gender-Based Count

Fig. 17. Analysis of glucose.



(a). Occupation Type Ratio (b). Count

Fig. 18. Analysis of occupation.



(A). Ratio of Residency Type (b). Count

Fig. 19. Analysis of residence type.

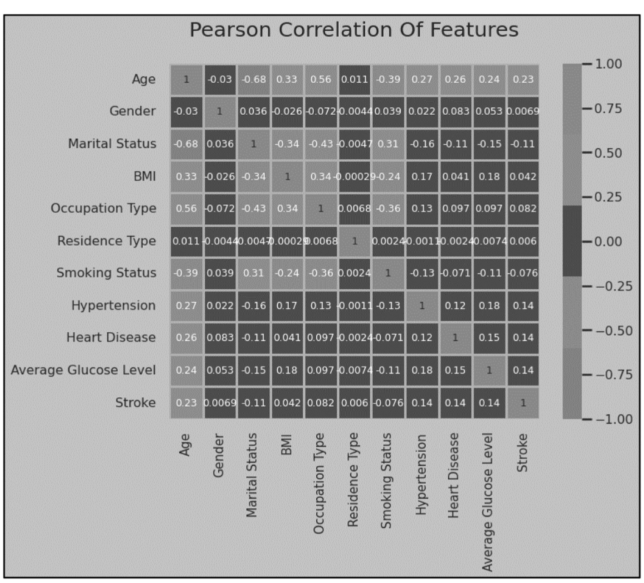


Fig. 20. Patient correlation type.

data. WSN includes multiple nodes, each with limited computational capabilities and memory. For the long-time process, this WSN should work hard to reach the remote location; this performance may ensure one extra challenge again.

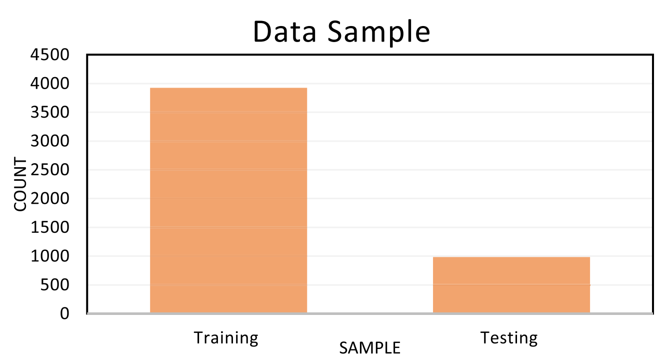


Fig. 21. Data sampling.

11. Conclusion

This paper demonstrates how WSN is used in medical industrial applications to monitor and analyze sensor-generated environmental and patient health-related data. Different types of sensor nodes are used to generate various kinds of data. Based on the data, the proposed Deep CNN and LSTM models automatically manage the medical industry application concerning the infrastructure, patients, doctors, and treatment. The WSN contributes to more sensors that can be deployed in the healthcare environment to monitor and record indoor and outdoor conditions. The major concern is monitoring the patient's health condition using medical sensors. The sensor data is analyzed using DCNN

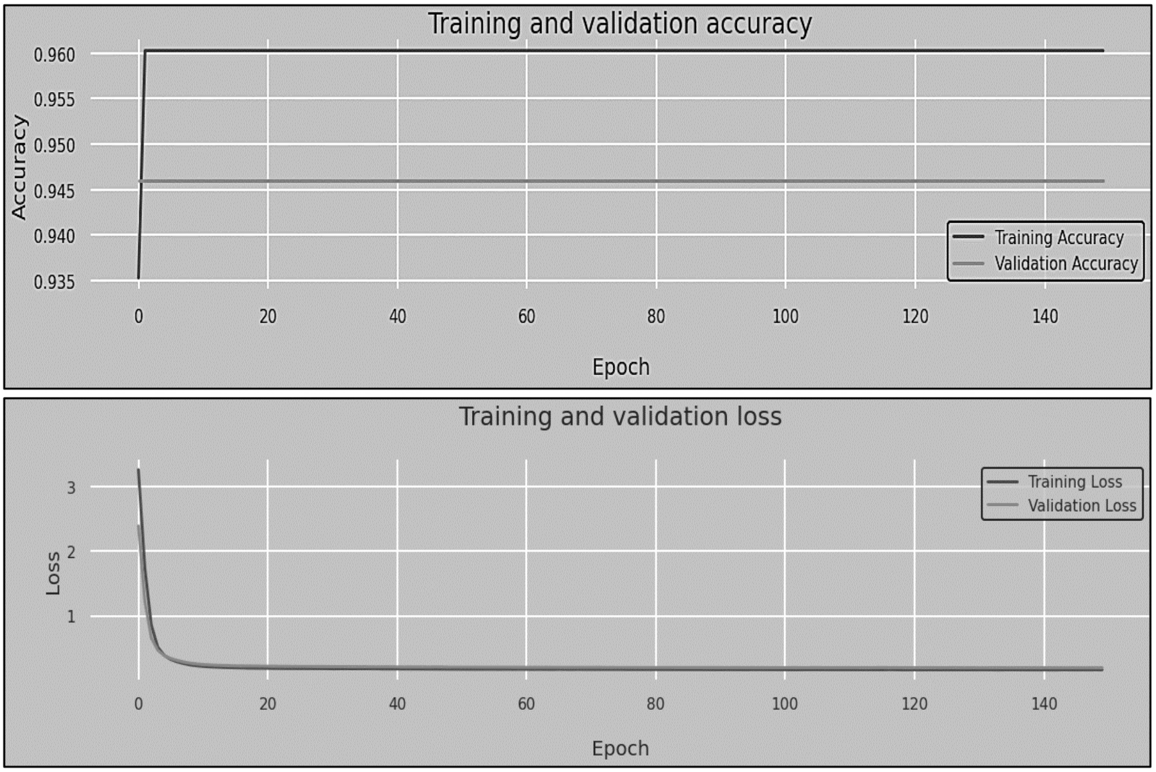


Fig. 22. Accuracy Vs. loss comparison of deep learning model.

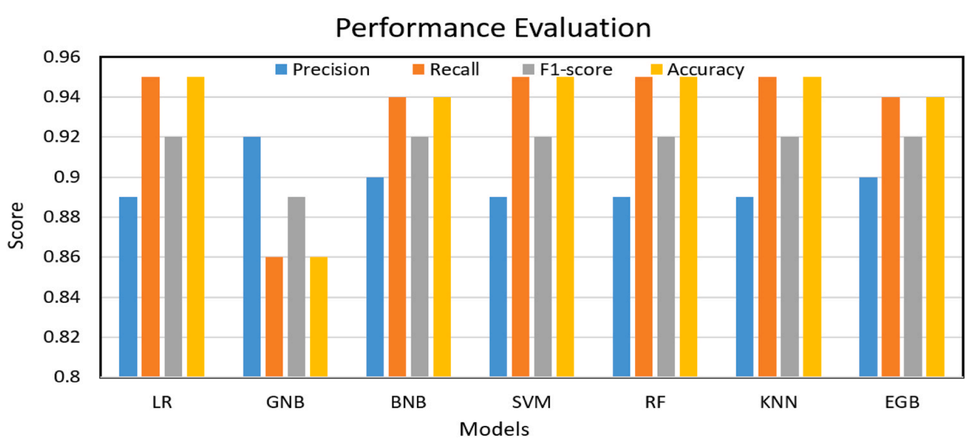


Fig. 23. Performance metrics result.

and LSTM models to predict abnormal conditions and diseases from various data, such as numeric, alpha-numeric, images, signals, and video. The final classification result is generated by evaluating multiple medical histories, such as BMI, HR, average glucose level, etc. Using NS2 simulation software, the final result of the model is predicted. The analysis indicates that the proposed deep learning model (CNN and LSTM) has achieved 96 % in classifying normal and abnormal patient results with a 0.08 loss rate. In addition, the model comparison result indicates that compared to another method, the combination of the DL model achieved a high accuracy of more than 5 %.

Author contributions

Writing and Experimentation, Technical Evaluation and Inference – All Authors.

Statements and declarations

The authors certify that this material or similar material has not been and will not be submitted to or published in any other publication before. Furthermore, the authors certify that they have participated sufficiently in the work to take public responsibility for the content, including participation in the concept, design, analysis, writing, or revision of the manuscript.

Compliance with ethical standards

Not applicable. No human and/or animal studies have been executed.

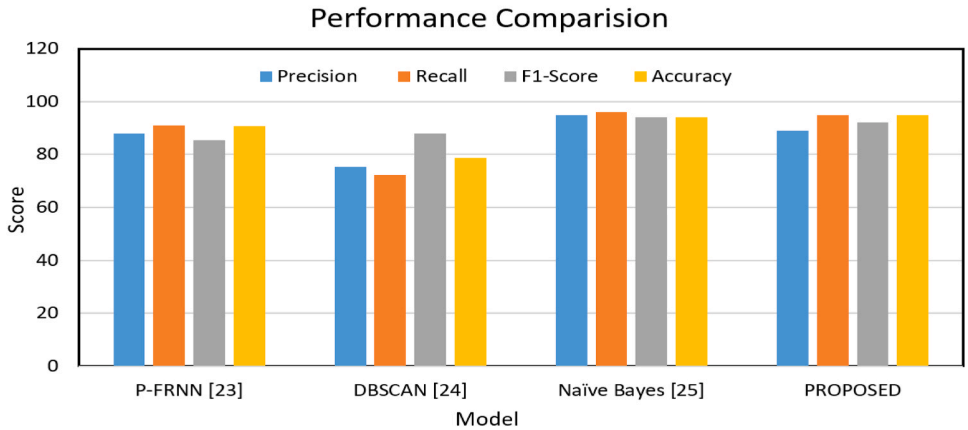


Fig. 24. Performance metrics comparison result.

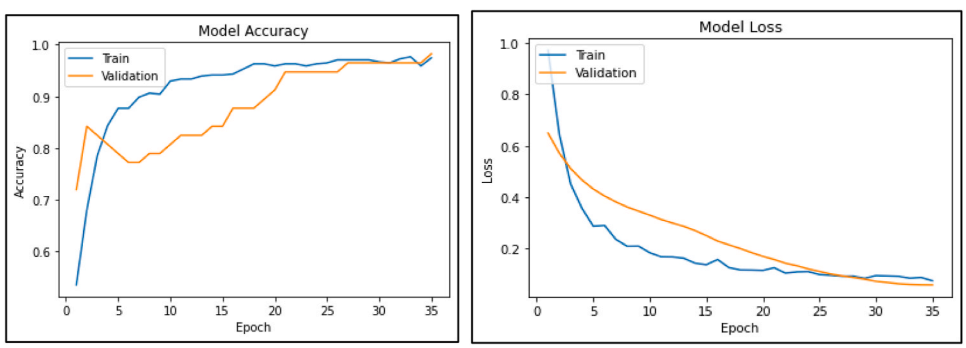


Fig. 25. Accuracy Vs. loss comparison of CNN.

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Informed consent statement

Not applicable.

CRediT authorship contribution statement

Alsowail Rakan A.: Writing – review & editing, Resources, Project administration, Methodology, Funding acquisition. **Pandiaraj Tharani:** Writing – review & editing, Validation, Investigation, Funding acquisition. **S Arunprakash:** Writing – original draft, Validation, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **R Manikandan:** Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The datasets used and analysed during the current study are available from the corresponding author on reasonable request.

References

[1] O. Chipara, C. Lu, T.C. Bailey, G.-C. Roman, Reliable patient monitoring: A clinical study in a step-down hospital unit. Technical Report WUCSE-2009-82, Department of Computer Science and Engineering, Washington University at St. Louis, Dec 2009.
[2] JeongGil Ko, JongHyun Lim, Yin Chen, Razvan Musaloiu-E, Andreas Terzis, Gerald Masson, Tia Gao, Walt Destler, Leo Selavo, Richard Dutton, MEDiSN: Medical Emergency Detection in Sensor Networks. ACM Transactions on Embedded Computing Systems (TECS), Special Issue on Wireless Health Systems, 2010.
[3] S. Kumar. Autosense. NIH GEI project at The University of Memphis. Available from: ⟨http://www.cs.memphis.edu/⟩˜santosh/AutoSense.html, 2007.
[4] K. Patrick. A tool for geospatial analysis of physical activity: Physical activity location measurement system (palms). NIH GEI project at the University of California, San Diego. Available from: ⟨http://www.gei.nih.gov/exposurebiology/⟩ program/docs/KevinPatrick.pdf, 2007.
[5] T. Gao, C. Pesto, L. Selavo, Y. Chen, J. Ko, J. Lim, A. Terzis, A. Watt, J. Jeng, B. Chen, et al., Wireless medical sensor networks in emergency response: implementation and pilot results, IEEE Int. Conf. Technol. Homel. Secur. (2008).
[6] David Malan, Thaddeus Fulford-Jones, Matt Welsh, and Steve Moulton. CodeBlue: An Ad Hoc Sensor Network Infrastructure for Emergency Medical Care. In Proceedings of the MobiSys 2004 Workshop on Applications of Mobile Embedded Systems (WAMES 2004), June 2004.
[7] G. Virone, A. Wood, L. Selavo, Q. Cao, L. Fang, T. Doan, Z. He, and J.A. Stankovic. An Advanced Wireless Sensor Network for Health Monitoring. In Proceeding of the Transdisciplinary Conference on Distributed Diagnosis and Home Healthcare (D2H2), April 2006.
[8] A. Darwish, A.E. Hassanien, Wearable and implantable wireless sensor network solutions for healthcare monitoring, Sens. (Basel) 11 (6) (2011) 5561–5595, https://doi.org/10.3390/s110605561. Epub 2011 May 26. Erratum in: Sensors (Basel). 2012;12(9):12375-6. PMID: 22163914; PMCID: PMC3231450.

[9] Harpinderjeet Kaur Sekhon and Vikramjit Singh, (2017), "CLUSTERING IN WIRELESS SENSOR NETWORK: A REVIEW", I n t e r n a t i o n a l j o u r n a l o f C o m p u t e r s a n d T e c h n o l o g y, Vol. 16, No. 6, pp. 6987-6993.
[10] K. Eghonghon Ukhurebor, I. Odesanya, S. Soo Tyokighir, R. George Kerry, A. Samson Olayinka, A. Oluwafemi Bobadoye, Wireless Sensor Networks: Applications and Challenges, IntechOpen, 2021, https://doi.org/10.5772/intechopen.93660.
[11] Humayun, A., Niaz, M., Umar, M., & Mujahid, M. (2017). Impact on the usage of wireless sensor networks in the healthcare sector. arXiv preprint arXiv: 1705.06021.
[12] T. Jabeen, I. Jabeen, H. Ashraf, N.Z. Jhanjhi, A. Yassine, M.S. Hossain, An intelligent healthcare system using IoT in a wireless sensor network, Sensors 23 (11) (2023) 5055.
[13] M.M. Islam, A. Rahaman, M.R. Islam, Development of smart healthcare monitoring system in IoT environment, SN Comput. Sci. 1 (2020) 185, https://doi.org/10.1007/s42979-020-00195-y.
[14] S. Mishra, A. Raj, A. Kayal, V. Choudhary, P. Verma, L. Biswal, Study of cluster based routing protocols in wireless sensor networks, Int. J. Sci. Eng. Res. 3 (7) (2012).
[15] D. Kandris, E. Anastasiadis, Advanced wireless sensor networks: applications, challenges and research trends, Electronics 13 (12) (2024) 2268.
[16] Basholli, F., Qorri, E., & Minga, J. Applications of wireless sensor networks in health care.
[17] A. Benfradj, A. Thaljaoui, T. Moulahi, R.U. Khan, A. Alabdulatif, P. Lorenz, Integrating artificial intelligence (AI) with sensor networks: trends, challenges, and future directions, J. King Saud. Univ. Comput. Inf. Sci. 36 (1) (2024) 101892.
[18] R. Kashyap, Applications of wireless sensor networks in healthcare. In IoT and WSN applications for modern agricultural advancements: Emerging research and opportunities, IGI Global, 2020, pp. 8–40.
[19] A. Tzounis, N. Katsoulas, T. Bartzanas, C. Kittas, Internet of Things in agriculture, recent advances and future challenges, Biosyst. Eng. 164 (2017) 31–48.
[20] N. Saleh, A. Kassem, A.M. Haidar, Energy-efficient architecture for wireless sensor networks in healthcare applications, IEEE Access 6 (2018) 6478–6486.
[21] A.N. Bahache, N. Chikouche, F. Mezrag, Authentication schemes for healthcare applications using wireless medical sensor networks: a survey, SN Comput. Sci. 3 (5) (2022) 382.
[22] M. Shakeri, A. Sadeghi-Niaraki, S.M. Choi, S.R. Islam, Performance analysis of IoT-based health and environment WSN deployment, Sensors 20 (20) (2020) 5923.
[23] H. Doshi, A. Shankar, Wireless Sensor Network Application for IoT-Based Healthcare System. In Data-Driven Approach Towards Disruptive Technologies: Proceedings of MIDAS 2020, Springer Singapore, 2021, pp. 287–307.
[24] Kaidi, H.M., Izhar, M.A.M., Dziyauddin, R.A., Shaiful, N.E., Ahmad, A. A comprehensive review on wireless healthcare monitoring: system components. IEEE Access. 1220243500835032.
[25] E. Aarthi, J.D. Daniel, G.M. Suba, N.P. Dharani, C.P. Devi, A naive bayes approach for improving heart disease detection on healthcare monitoring through IoT and WSN, Int. J. Intell. Syst. Appl. Eng. 12 (2s) (2024) 553–570.
[26] M.Y.B. Murthy, A. Koteswararao, A detailed review of applications, merits, and demerits of WSN with IoT, Int. J. Auton. Adapt. Commun. Syst. 17 (1) (2024) 68–88.
[27] R. Krishnamoorthy, M. Gupta, G. Swathi, K. Tanaka, C. Raja, J.N.V. Ramesh, An intelligent IoT-BAsed Smart Healthcare Monitoring System Using Machine Learning, In 5G-Based Smart Hospitals and Healthcare Systems, CRC Press (2024) 230–247.
[28] C.M. Ardila, A.M. Vivares-Builes, Artificial intelligence through wireless sensors applied in restorative dentistry: a systematic review, Dentistry J 12 (5) (2024) 120.
[29] K.K. Saini, D. Kaur, D. Kumar, B. Kumar, An efficient three-factor authentication protocol for wireless healthcare sensor networks, Multimed. Tools Appl. (2024) 1–23.
[30] H. Sharma, A. Haque, F. Blaabjerg, Machine learning in wireless sensor networks for smart cities: a survey, Electronics 10 (9) (2021) 1012.
[31] N.B. Oğur, M. Al-Hubaishi, C. Çeken, IoT data analytics architecture for smart healthcare using RFID and WSN, ETRI 44 (1) (2022) 135–146.
[32] Upreti, K., Kumar, N., Alam, M.S., Verma, A., Nandan, M., Gupta, A.K. Machine learning-based congestion control routing strategy for healthcare IoT-enabled wireless sensor networks (September) 2021 Fourth International Conference on Electrical, Computer and Communication Technologies (ICECCT), IEEE, 2021, pp. 1–6 (September).
[33] S. Balakrishnan, K. Suresh Kumar, L. Ramanathan, et al., IoT for health monitoring system based on machine learning algorithm, Wirel. Pers, Commun 124 (2022) 189–205.
[34] J.B. Awotunde, S.O Folorunso, A.K. Bhoi, P.O. Adebayo, M.F. Ijaz, Disease diagnosis system for IoT-based wearable body sensors with a machine learning algorithm, Hybrid Artif. Intell. IoT Healthc. (2021) 201–222.
[35] M. Pundir, J.K. Sandhu, A systematic review of quality of service in wireless sensor networks using machine learning: recent trend and future vision, J. Netw. Comput. Appl. 188 (2021) 103084.
[36] G. Gardašević, K. Katzis, D. Bajić, L. Berbakov, Emerging wireless sensor networks and Internet of Things technologies—Foundations of smart healthcare, Sensors 20 (13) (2020) 3619.
[37] H. Aldabbas, D. Albashish, K. Khatatneh, R. Amin, An architecture of IoT-aware healthcare smart system by leveraging machine learning, Int. Arab J. Inf. Technol. 19 (2) (2022) 160–172.
[38] B.M. Elbagoury, L. Vladareanu, V. Vlădăreanu, A.B. Salem, A.M. Travediu, M. I. Roushdy, A hybrid stacked CNN and residual feedback GMDH-LSTM deep learning model for stroke prediction was applied on a mobile AI smart hospital platform, Sensors 23 (7) (2023) 3500.
[39] R. Lavanya, D. Vidyabharathi, S.S. Kumar, M. Mali, M. Arunkumar, S.S. Aravinth, M. Tesfayohanis, [Retracted] wearable sensor-based edge computing framework for cardiac arrhythmia detection and acute stroke prediction, J. Sens. 2023 (1) (2023) 3082870.
[40] Elbagoury, B.M., Maskeliuans, R., Zaghow, M., Kamel, N. A Novel Inception Deep Learning Model for Mobile AI Stroke Prediction System: AI Research and Industry Perspectives for Connected Health (March) In 2024 6th International Conference on Computing and Informatics (ICCI), IEEE, 2024, pp. 366–376 (March).
[41] F. Jin, M. Zou, X. Peng, H. Lei, Y. Ren, Deep learning-enhanced internet of things for activity recognition in post-stroke rehabilitation, IEEE J. Biomed. Health Inform. (2023).
[42] W.C. Lai, L. Guo, K. Bialkowski, A. Bialkowski, An explainable deep learning method for microwave head stroke localization, IEEE J. Electromagn., RF Microw. Med. Biol. (2023).
[43] G. Rajkumar, T.G. Devi, A. Srinivasan, Heart disease prediction using IoT-based framework and improved deep learning approach: medical application, Med. Eng. Phys. 111 (2023) 103937.
[44] B. Ge, H. Yang, P. Ma, T. Guo, J. Pan, W. Wang, Detection of pulmonary hypertension associated with congenital heart disease based on time-frequency domain and deep learning features, Biomed. Signal Process. Control 81 (2023) 104316.
[45] B.S. Deepapriya, P. Kumar, G. Nandakumar, S. Gnanavel, R. Padmanaban, A. K. Anbarasan, K. Meena, Performance evaluation of deep learning techniques for lung cancer prediction, Soft Comput. 27 (13) (2023) 9191–9198.
[46] L. Luo, X. Wang, Y. Lin, X. Ma, A. Tan, R. Chan, H. Chen, Deep learning in breast cancer imaging: a decade of progress and future directions, IEEE Rev. Biomed. Eng. (2024).
[47] K.R. Pedada, B. Rao, K.K. Patro, J.P. Allam, M.M. Jamjoom, N.A. Samee, A novel approach for brain tumour detection using deep learning-based technique, Biomed. Signal Process. Control 82 (2023) 104549.
[48] M.M. Arslan, X. Yang, Z. Zhang, S.U. Rahman, M. Ullah, Q.H. Abbasi, Advancing healthcare monitoring: integrating machine learning with innovative wearable and wireless systems for comprehensive patient care, IEEE Sens. J. (2024).
[49] Y. Qiu, L. Ma, R. Priyadarshi, Deep learning challenges and prospects in wireless sensor network deployment, Arch. Comput. Methods Eng. (2024) 1–24.
[50] A.K. Jameil, H. Al-Raweshidy, Implementation and evaluation of digital twin framework for Internet of Things-based healthcare systems, IET Wirel. Sens. Syst. (2024).
[51] P. Saini, R. Ahuja, V. Sai, Wireless sensor networks and iot revolutionizing healthcare: advancements, Applications, and Future Directions, in: Emerging Technologies and the Application of WSN and IoT, CRC Press, 2024, pp. 228–250.
[52] G.C. Iannelli, F. Dell'Acqua, Extensive exposure mapping in urban areas through deep analysis of street-level pictures for floor count determination, Urban Sci. 1 (2) (2017) 16, https://doi.org/10.3390/urbansci1020016.